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Orchestrating AI Agents in Spec-Driven Development with Kiro IDE



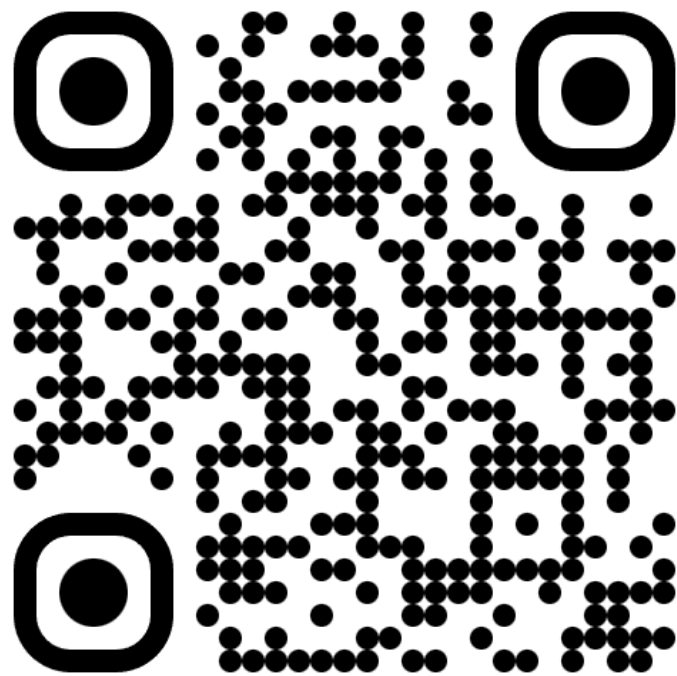
Haim Michael

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Orchestrating AI Agents in Spec-Driven Development with Kiro IDE

Haim Michael

Founder & CEO @ Zindell Technologies
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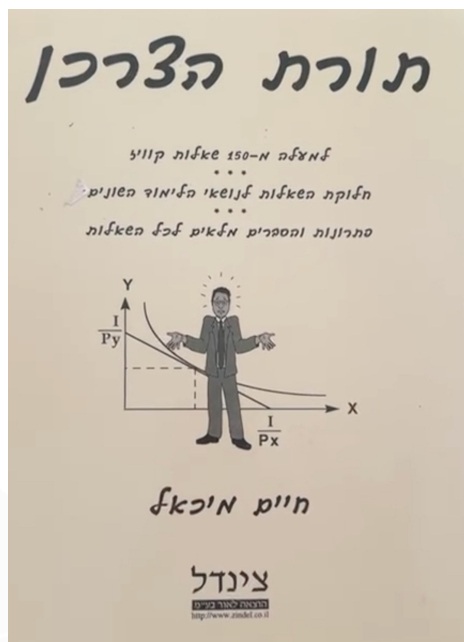
About Me

My name is Haim Michael, and I live in Tel-Aviv, Israel.

I have been into programming since my childhood. I have extensive experience with Java (30+ yrs), JavaScript (30+ yrs), C# (20+ yrs), Python (15+ yrs), PHP (20+ yrs), Scala (16+ yrs), and Kotlin (12+ yrs).

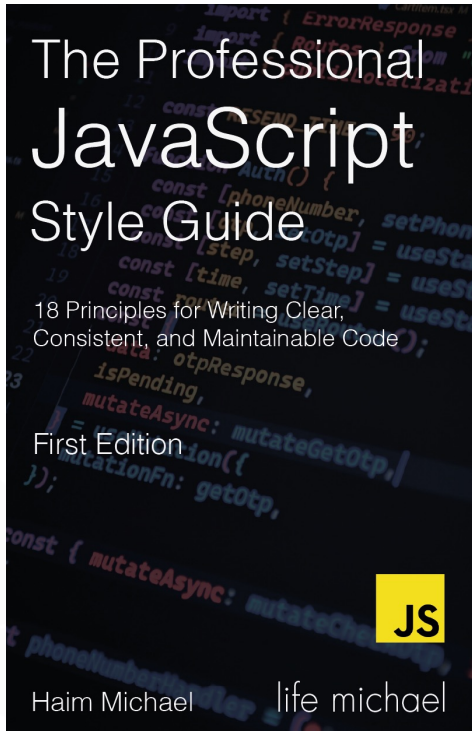
My passion lies in teaching and lifelong learning.

My Books

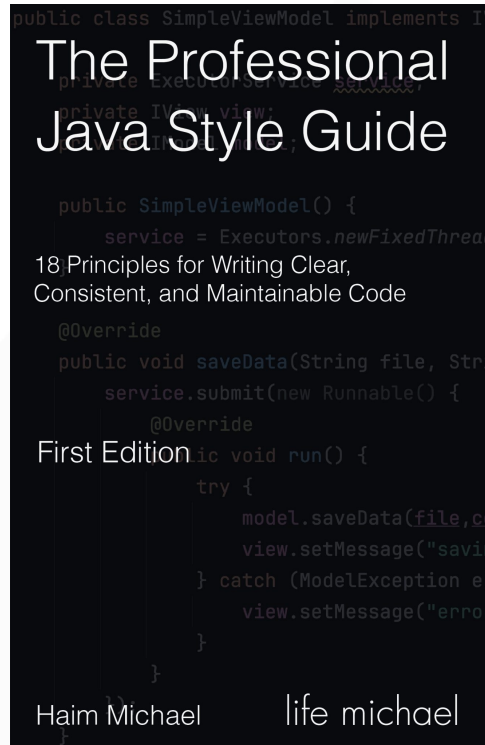
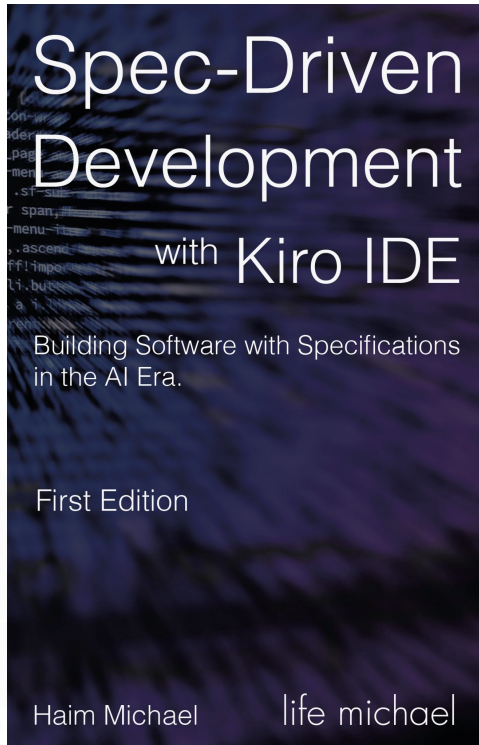


Teaching Micro Economic courses in the Open University

My Books



My Books



My Courses

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הרשמה לקורס

Courses (in Hebrew) @ <https://academy.lifemichael.com/he>

My Courses



Lambda Expressions in Kotlin

Acquire Practical Experience and In-depth Understanding!

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Course ★ 4.6 12 ratings

1.5 total hours 10 lectures

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Inner Classes in Java

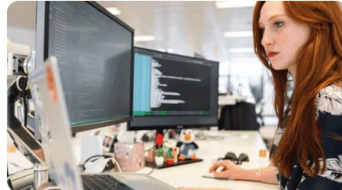
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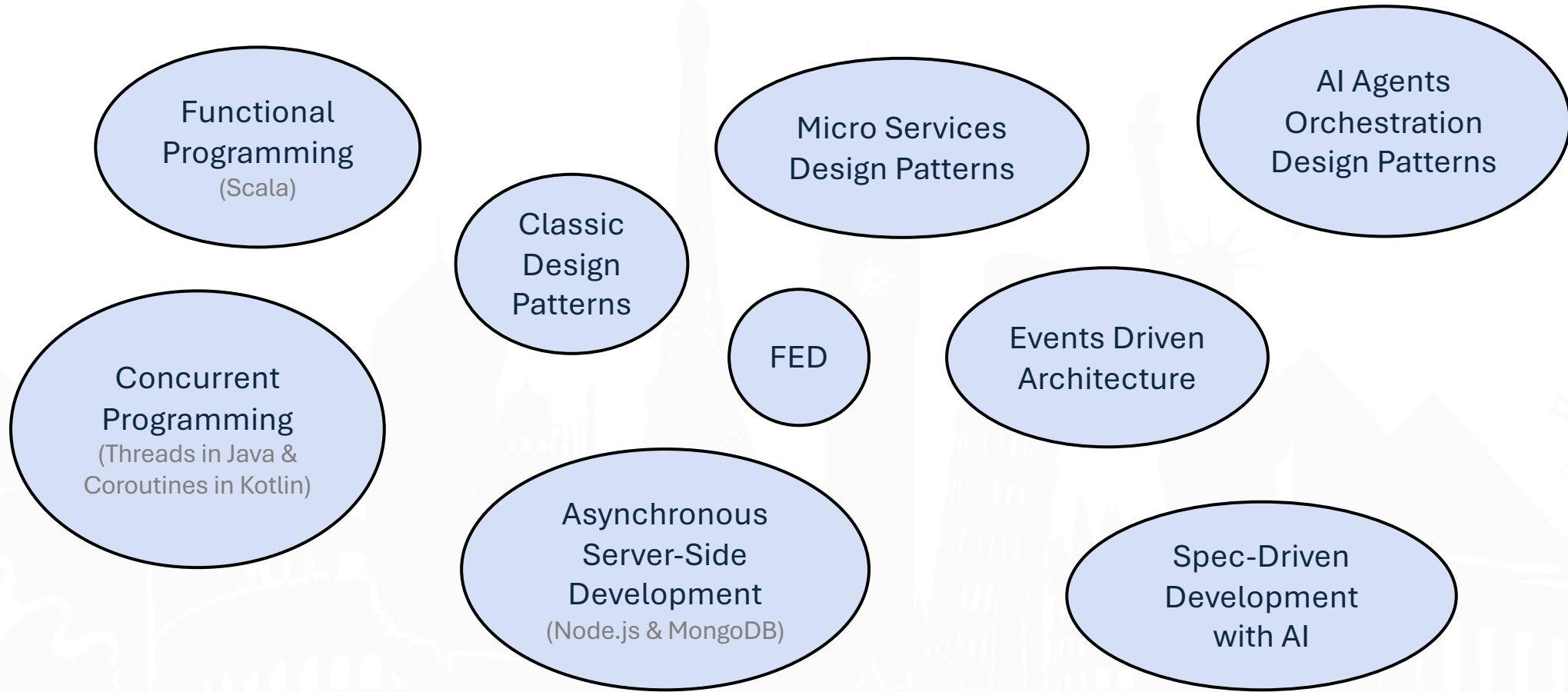
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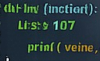
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The English Programming Language



The End of Code First Development

The large-scale code generation shifts the bottleneck from code generation to understanding and control.



```

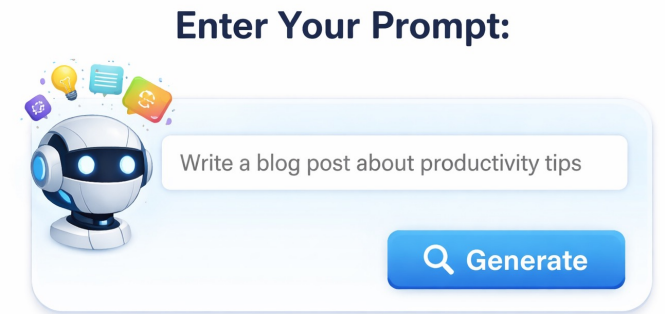
1 struct Node {
2     int value;
3     Node* next;
4 };
5
6 int main() {
7     Node* head = new Node(107);
8     head->next = new Node(108);
9     head->next->next = new Node(109);
10    return 0;
11 }

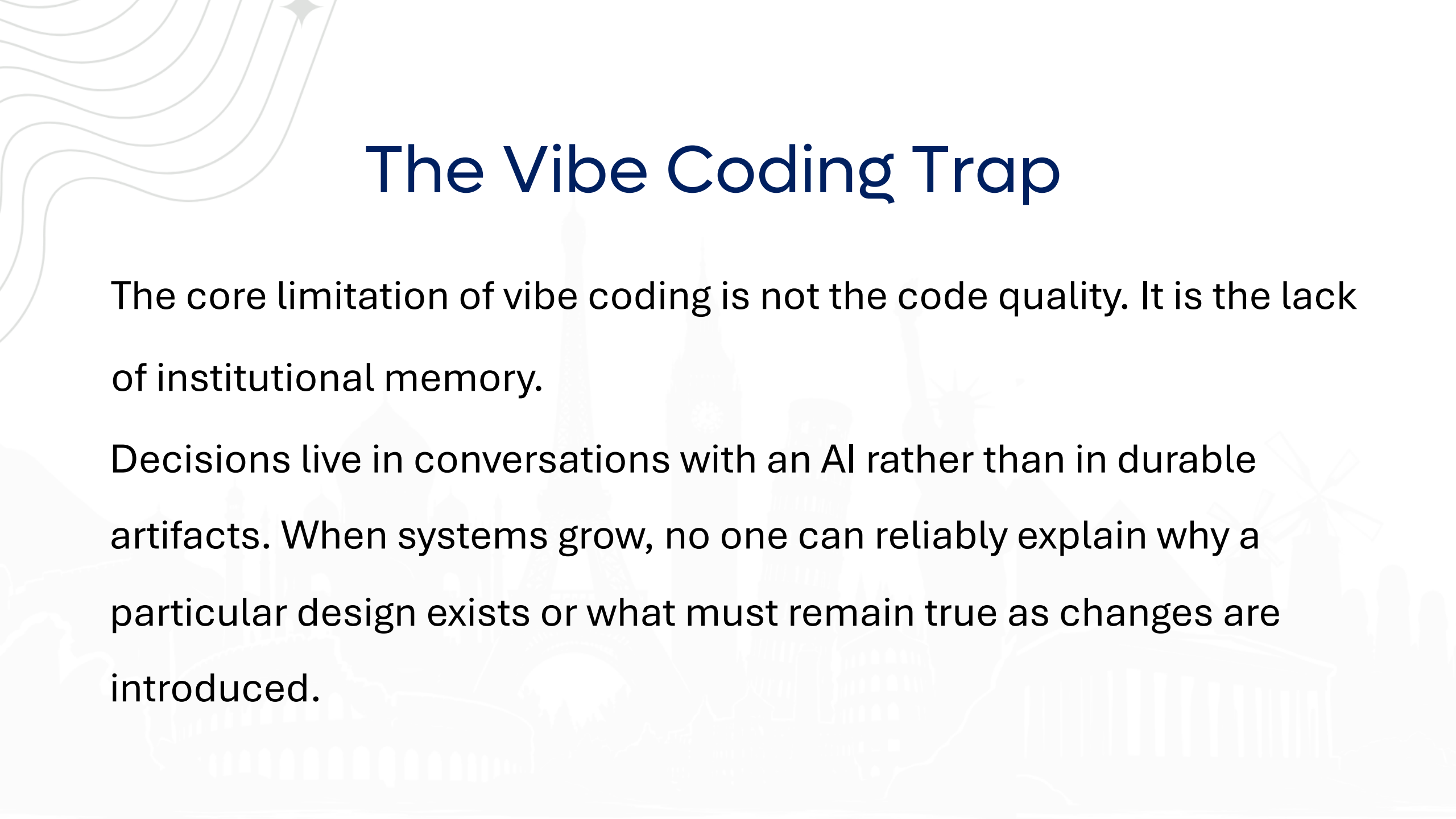
```

un lien en mémoire

The Prompt Driven Problem

The prompts cannot be managed or traced. When the systems evolve, intent is lost, and the behavior becomes difficult to explain or reproduce.





The Vibe Coding Trap

The core limitation of vibe coding is not the code quality. It is the lack of institutional memory.

Decisions live in conversations with an AI rather than in durable artifacts. When systems grow, no one can reliably explain why a particular design exists or what must remain true as changes are introduced.

The Vibe Coding Value

Vibe coding optimizes for exploration and immediacy. Professional software engineering optimizes for reliability, traceability, and long-term value.



Spec-Driven Development



Specs as the Missing Layer

Specs capture what the system must do, what it must never do, and under which conditions behavior changes. They translate human intent into a form that can be reviewed, versioned, validated, and shared. Both by humans and by machines..



Spec-Driven Development AI Tools

Tool	Type	Spec-Driven Depth	Role in SDD Stack	★ Rating
Kiro	AI IDE	Full	Spec = control layer	★★★★★★
Spec-Kit / BMAD	Frameworks	Very strong	Spec governance	★★★★★☆
OpenSpec	CLI	Moderate-strong	Lightweight spec layer	★★★★★☆
Intent	Agent platform	Strong	Spec + orchestration	★★★★★☆
Claude Code	Agentic coding system	Moderate (indirect)	Execution engine (with optional specs)	★★★★★☆
Cursor	AI IDE	Partial	Execution + speed	★★★★☆☆

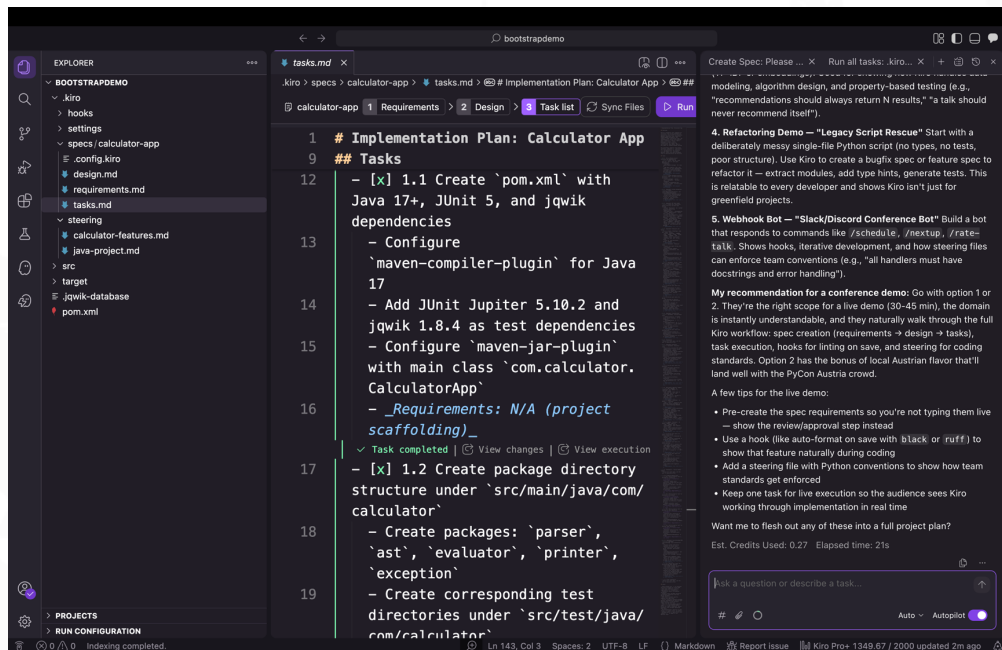
Created with ChatGPT. The prompt: Create a table that compares all the AI tools that support Spec-Driven Development.

The Kiro IDE

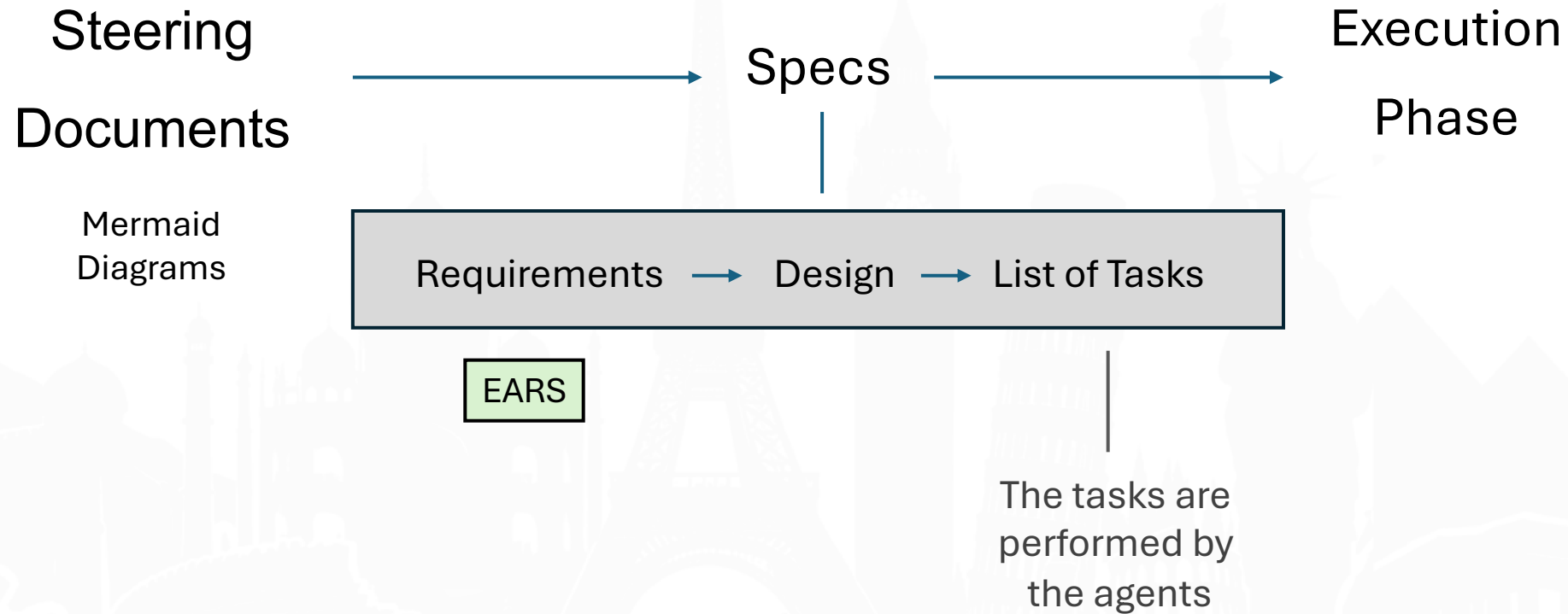


The Kiro IDE

The Kiro IDE was developed by Amazon and is based on VS Code. Kiro is an Agentic IDE.



The Development Process



AI Agents Hooks



What are Agent Hooks?

Agent hooks are automated triggers that execute predefined agent prompts or shell commands when specific events occur in your IDE. Rather than manually asking for routine tasks to be performed, hooks set up automated responses to events such as:

Linting

Updating Other Files

Static Analysis Tools

Updating Related Files

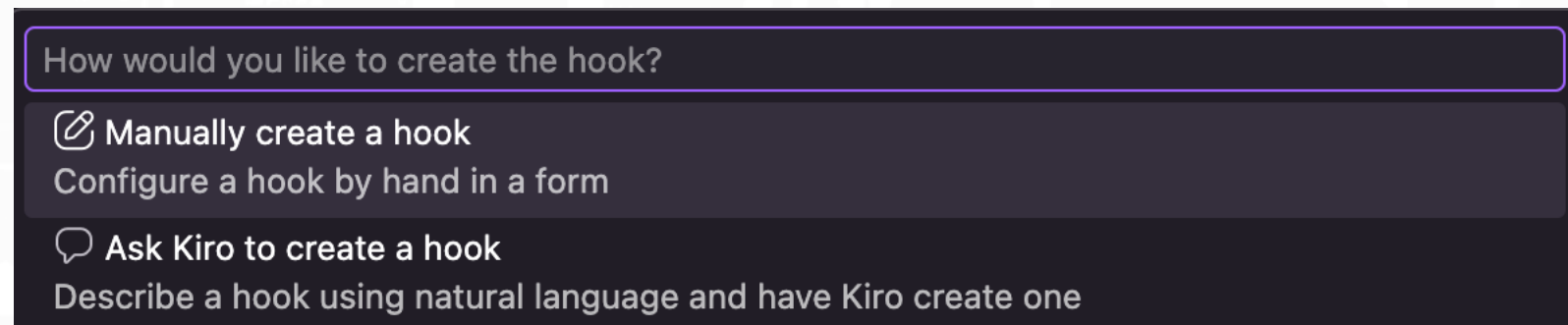
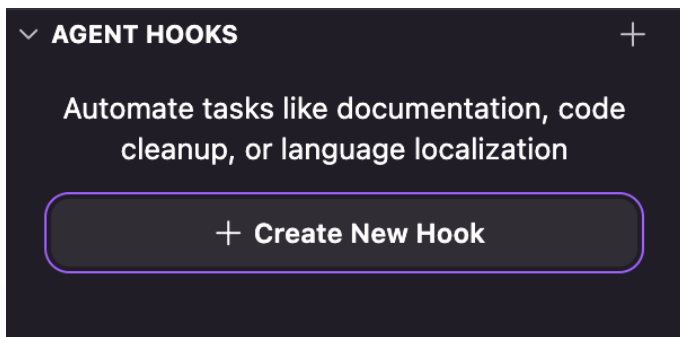
Create Tests

Code Formatting

Generating Documentation

Agent Hooks

There are two ways to create new hooks. Either you create them manually or you ask Kiro chatbot to create them for you.



Creating Simple Hook

Whenever a file is saved carefully review the code to ensure that it adheres every possible security best practices.

#



Auto ▾

Autopilot



Including Steering Documents

backend.md

categories.md

coding-standards.md

database.md

frontend.md

project-overview.md

python-crawler.md

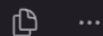
security.md

> Created Hook

Security Review on Save

Done. The hook is now active — whenever you save a `.php`, `.js`, `.py`, or `.html` file, it will trigger a security review checking for SQL injection, XSS, input validation, CSRF, error handling, secure headers, path traversal, command injection, session security, API key exposure, and HTTPS enforcement.

Est. Credits Used: 0.22 Elapsed time: 13s



Creating Simple Hook

 security-review-save

Hook enabled ☒

Title

Security Review on Save

Description

Whenever a file is saved, carefully review the code to ensure it adheres to every possible security best practice including input validation, SQL injection prevention, XSS prevention, CSRF protection, proper error handling, and secure configurations.

Creating Simple Hook

Event

File Saved ▾

File path(s) to watch

*.php ×

*.js ×

*.py ×

*.html ×

Action

Ask Kiro

Run Command

Provides instructions to the agent.

Instructions for Kiro agent

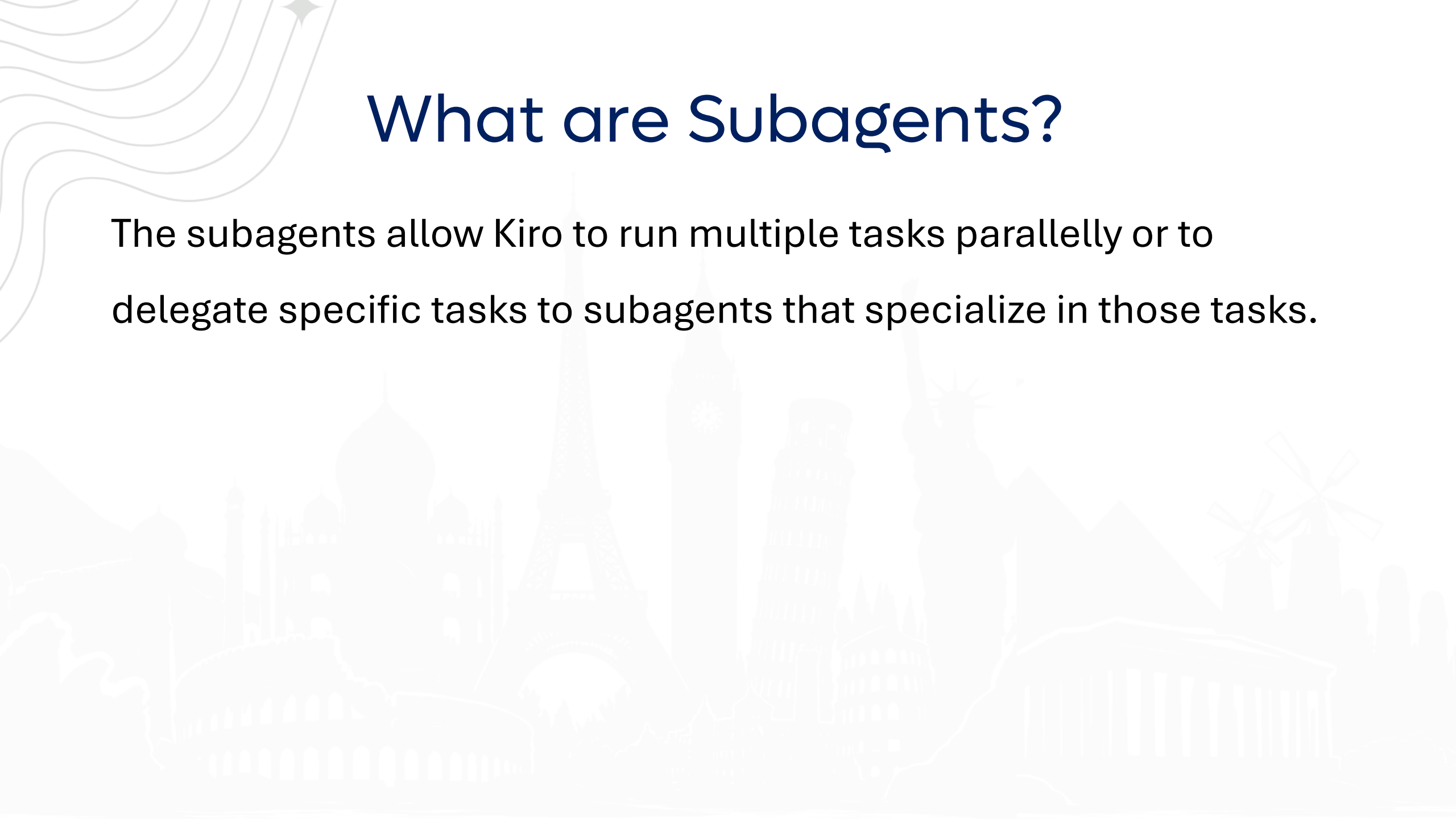
Carefully review the saved file for security best practices. Check for: 1) SQL injection vulnerabilities (ensure prepared statements are used), 2) XSS vulnerabilities (ensure proper output encoding, use `textContent` over `innerHTML`), 3) Input validation (reject suspicious characters, validate all user input), 4) CSRF protection, 5) Proper error handling (no internal details exposed to users), 6) Secure headers, 7) Path traversal prevention, 8) Command injection prevention, 9) Secure session

Subagents



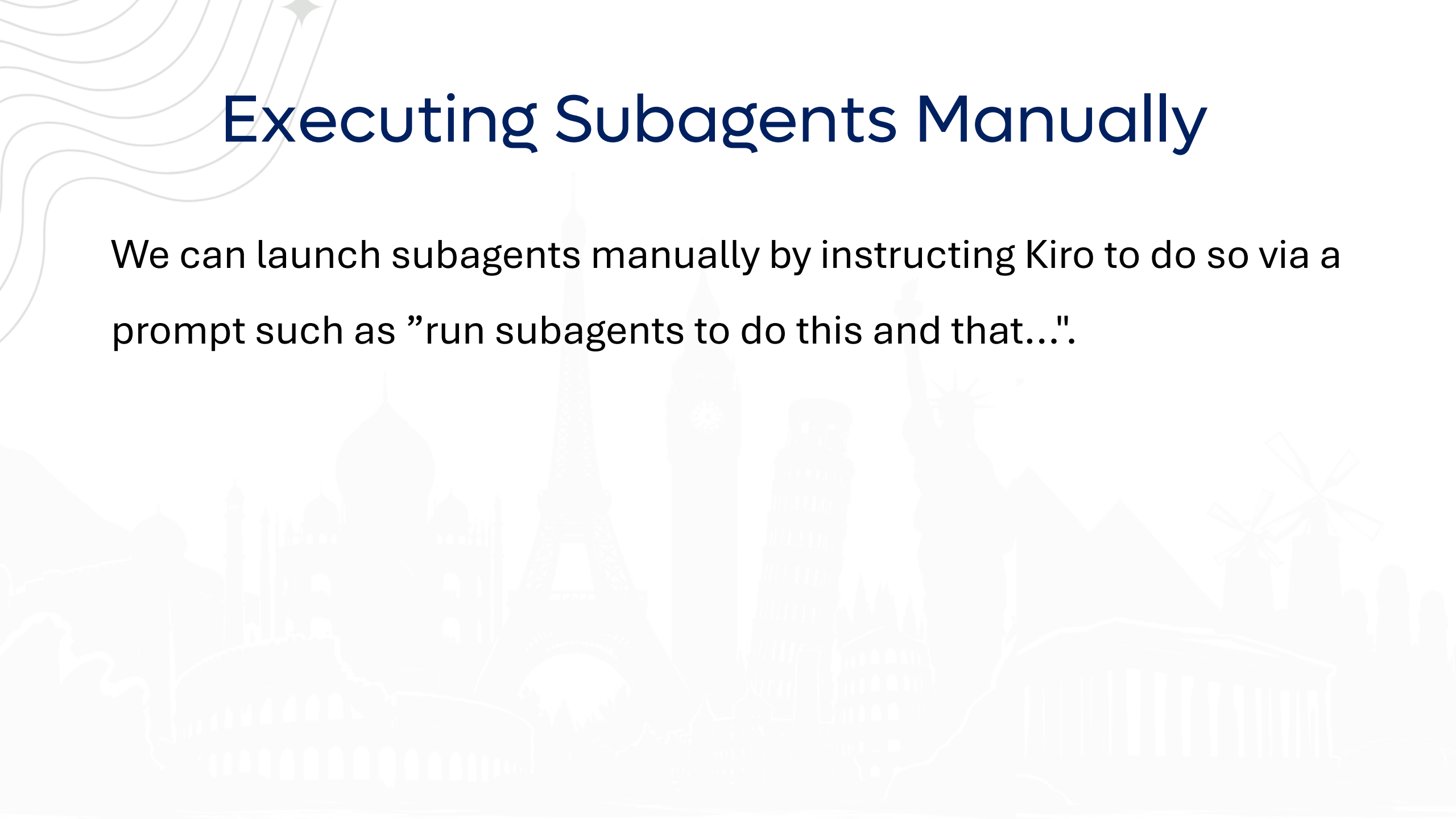
What are Subagents?

The subagents allow Kiro to run multiple tasks parallelly or to delegate specific tasks to subagents that specialize in those tasks.



Executing Subagents Manually

We can launch subagents manually by instructing Kiro to do so via a prompt such as "run subagents to do this and that...".



Executing Subagents Manually

1st Agent

You are a Refactoring Agent.
Improve code quality without changing observable behavior.
Focus on:

- Smaller functions
- Better naming
- Removing duplication
- Reducing coupling
- Improving readability
- Preserving public APIs
- Preserving tests
- Avoiding unnecessary abstractions

Before modifying code:

1. Identify the refactoring goal
2. Identify risky areas
3. Explain the expected behavior to preserve

After modifying code:

1. Summarize changes
2. Mention behavior-preservation assumptions
3. Recommend tests to run

2nd Agent

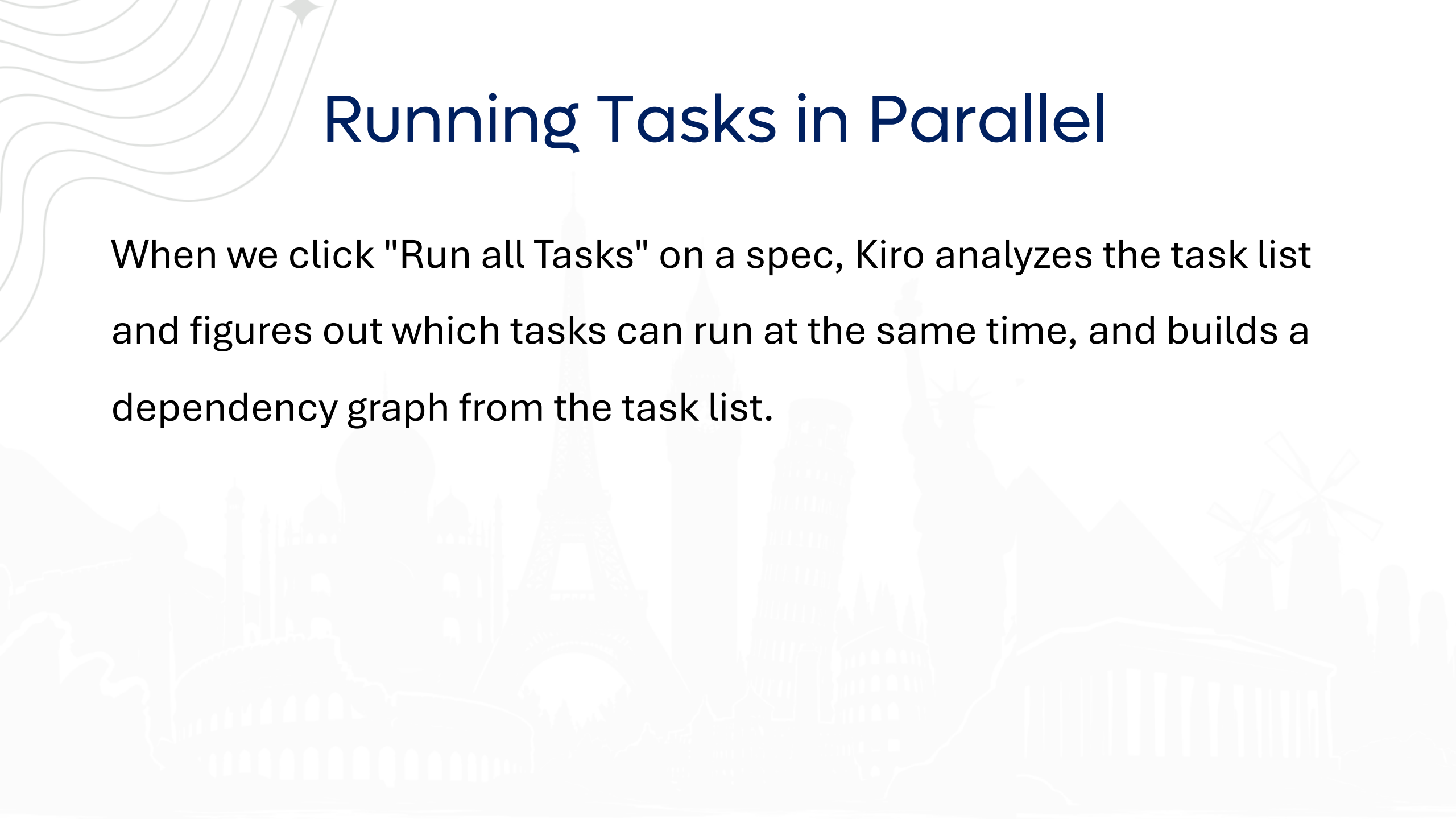
create a sub agent that is a Security Review Agent.
Review the codebase for security issues.
Focus on:

- Input validation
- Output escaping
- SQL injection
- XSS
- CSRF
- Authentication bypass
- Authorization errors
- Secrets in source code
- Unsafe file operations
- Unsafe shell execution
- Dangerous PHP functions
- WordPress-specific security issues

Do not rewrite the entire codebase.
Return only actionable findings with file names, line references if possible, risk level, and recommended fix.

Tasks in Parallel

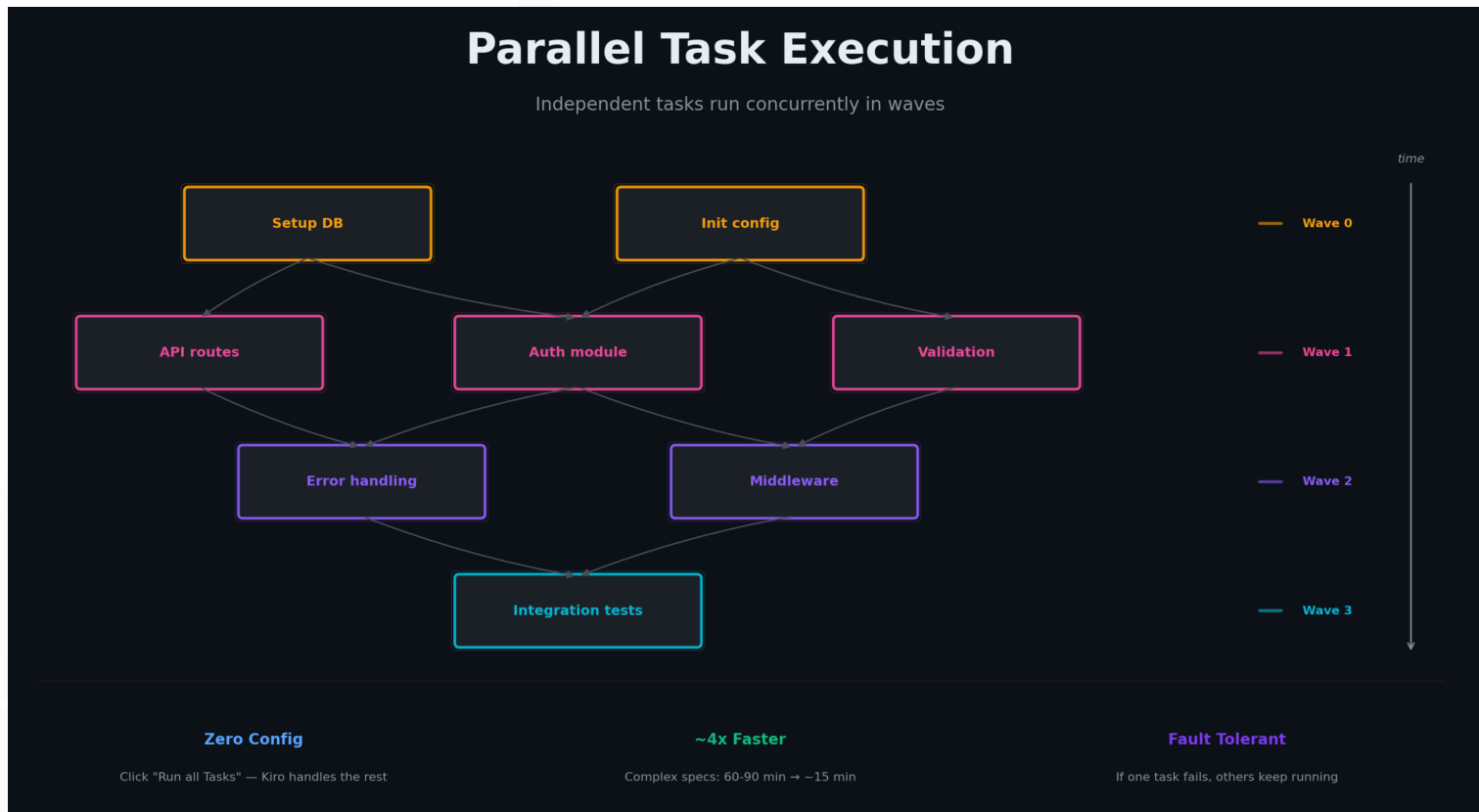




Running Tasks in Parallel

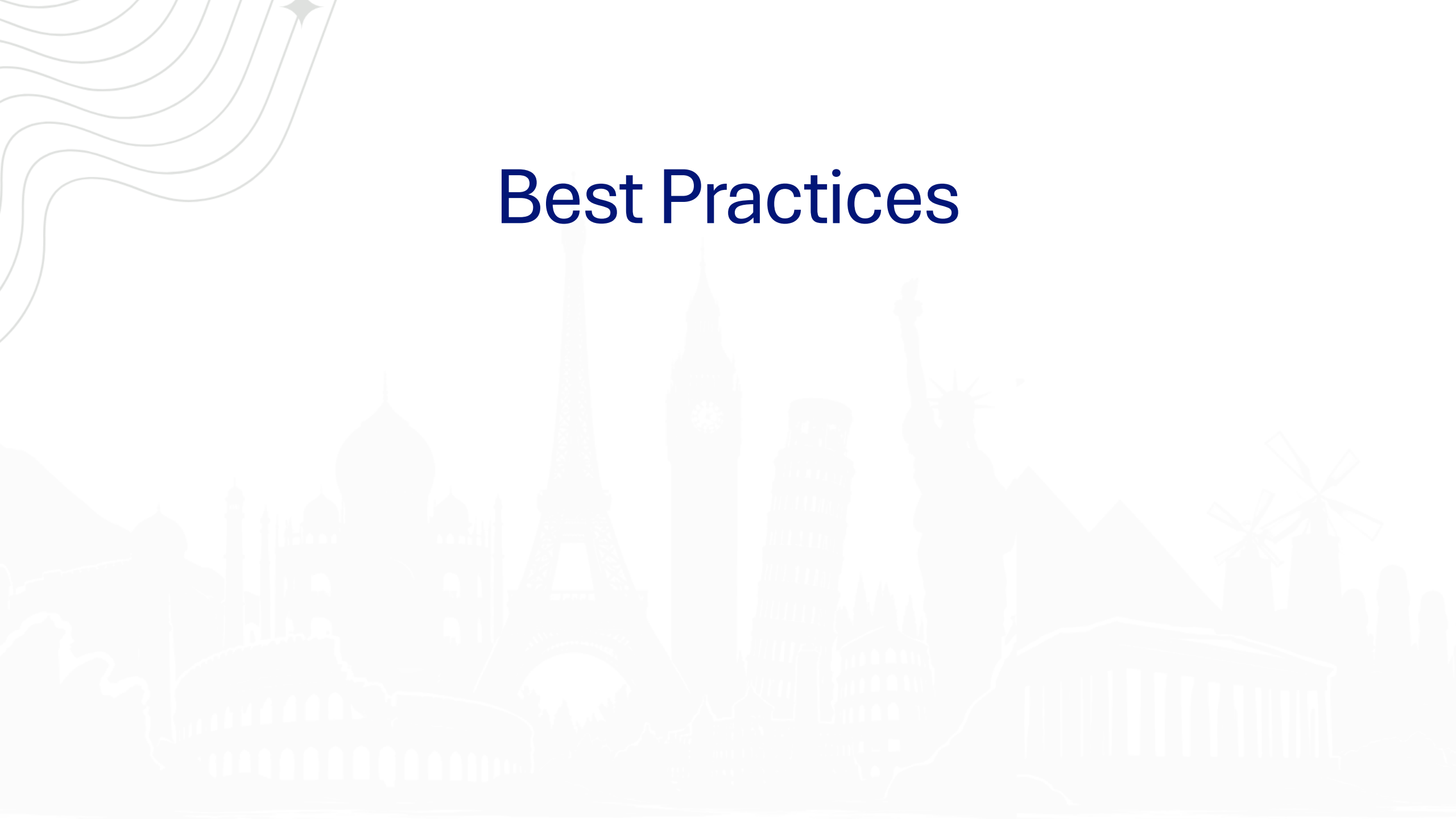
When we click "Run all Tasks" on a spec, Kiro analyzes the task list and figures out which tasks can run at the same time, and builds a dependency graph from the task list.

Running Tasks in Parallel



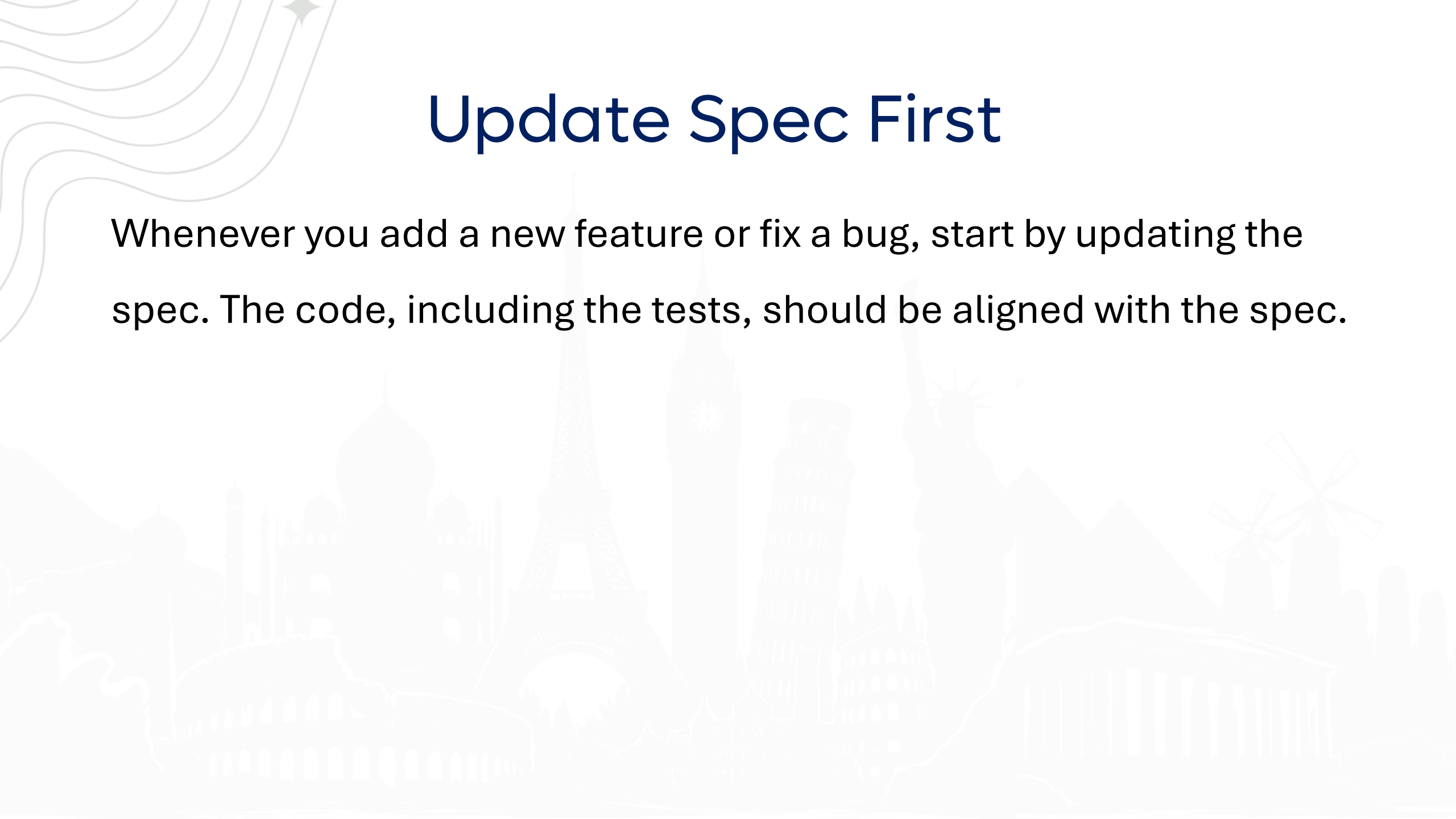
Taken from <https://kiro.dev/blog/faster-smarter-specs/>

Best Practices



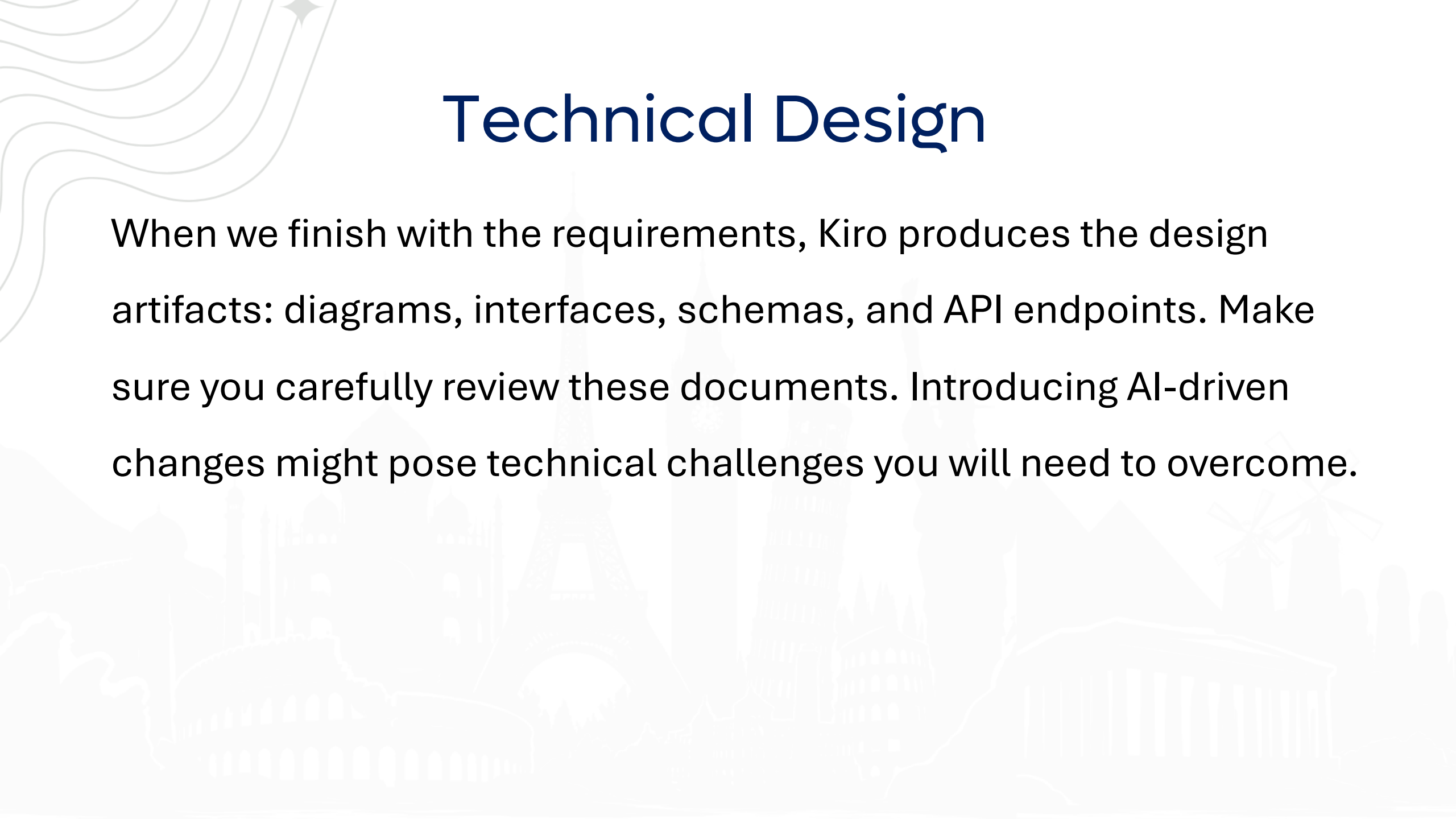
Update Spec First

Whenever you add a new feature or fix a bug, start by updating the spec. The code, including the tests, should be aligned with the spec.



Detailed Steering Documents

When having detailed steering documents, the steps that follow will produce better results. By doing so, you reduce re-work, backtracking, and misalignment across humans and the AI.



Technical Design

When we finish with the requirements, Kiro produces the design artifacts: diagrams, interfaces, schemas, and API endpoints. Make sure you carefully review these documents. Introducing AI-driven changes might pose technical challenges you will need to overcome.

AI Optimization

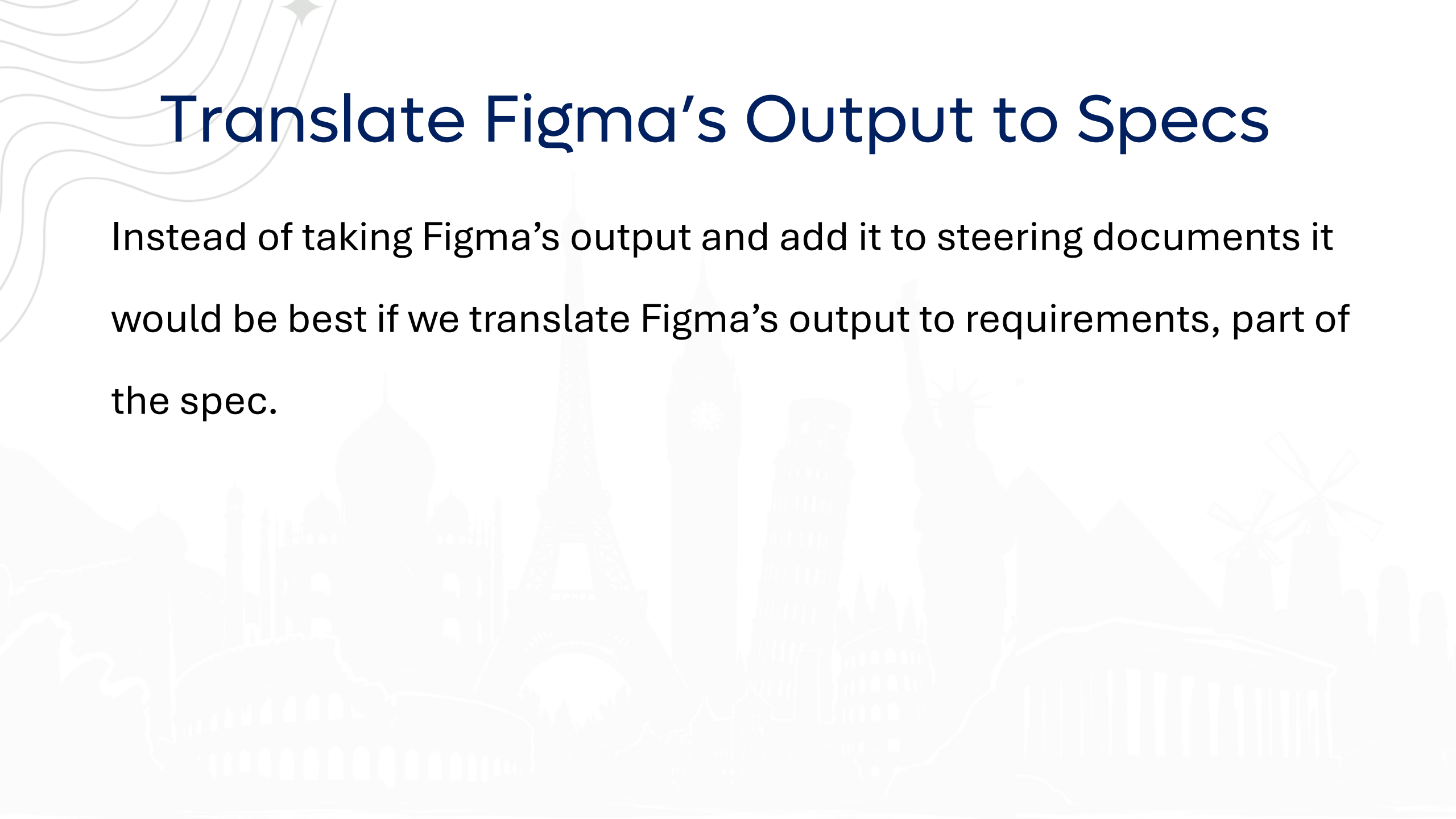
Make sure to configure the relevant MCP servers and to install the skills and powers that will assist AI.

Powers bundle MCP tools with knowledge and workflows. They activate dynamically based on context.

Skills are portable instruction packages that bundle instructions, scripts, and templates, and Kiro can activate them when relevant to your task.

Translate Figma's Output to Specs

Instead of taking Figma's output and add it to steering documents it would be best if we translate Figma's output to requirements, part of the spec.



Zero Human Companies



What are Zero Human Companies?

The core business operations, including the service (or product) development, are performed primarily by AI agents, while humans define the high-level intent, constraints, and strategic direction.

Diminishing Marginal Productivity

When additional units of a variable input (e.g., labor, compute, or effort) are added to a fixed set of other inputs, the incremental output (marginal productivity) produced by each additional unit will eventually decrease.



The Good Enough Approach

When having a less crowded working environment, it is easier to move forward when it is good enough.



The Future



Coding is Cheap

The software developer role is moving to a new higher level of abstraction. The possible paths include the following:

Software Builders (T Shaped People)

Software Architects (Server-Side)

Technologies Experts (Side by Side with Builders)

Becoming a software builder is a path that is available for many others, including product managers, UI/UX experts and especially those that are experts in their domain.

Tsunami of AI Based Services

The upcoming future seems to be dominated with many new AI based services and products.





Questions & Answers

Thanks for attending my talk :)

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